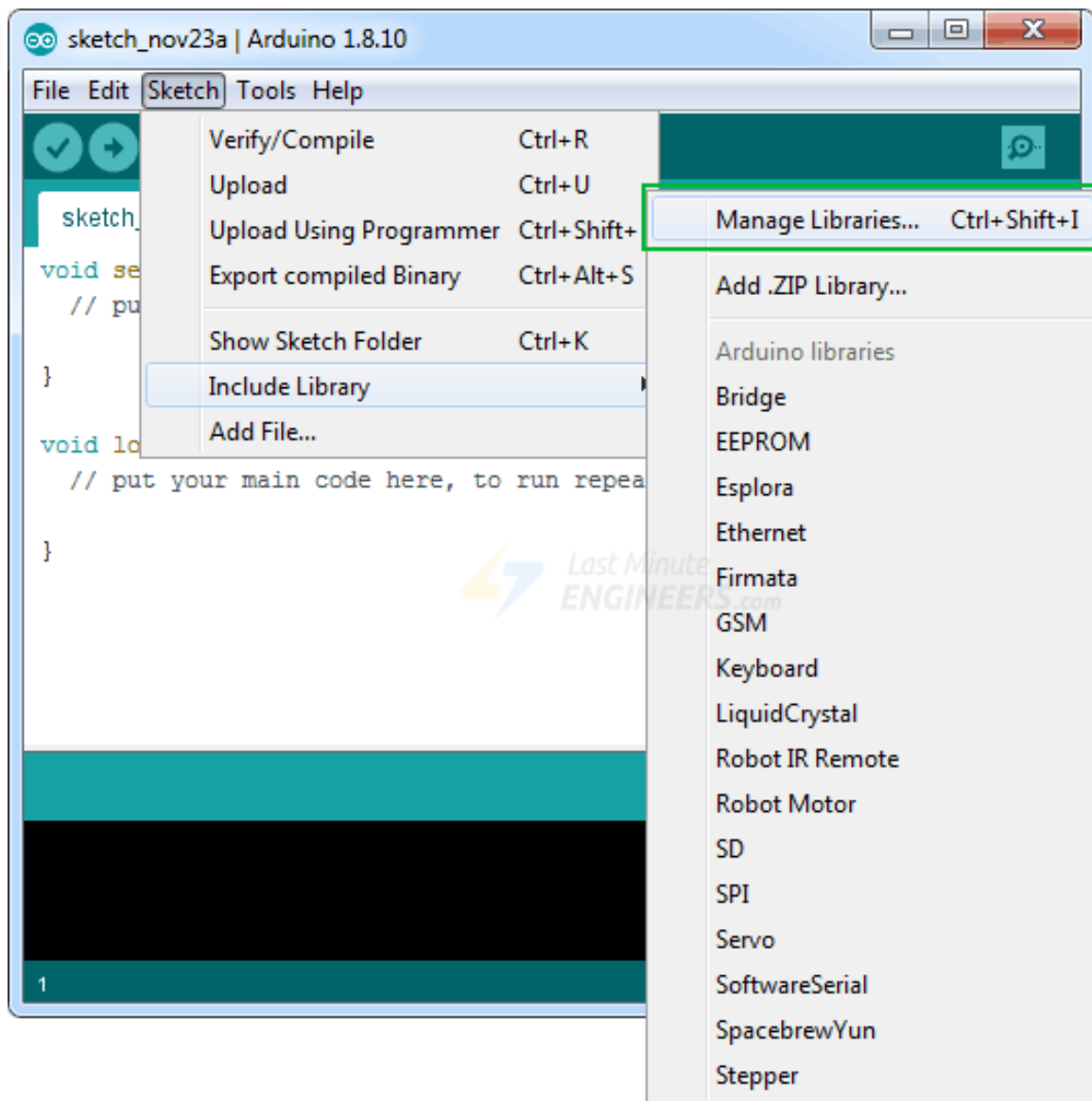


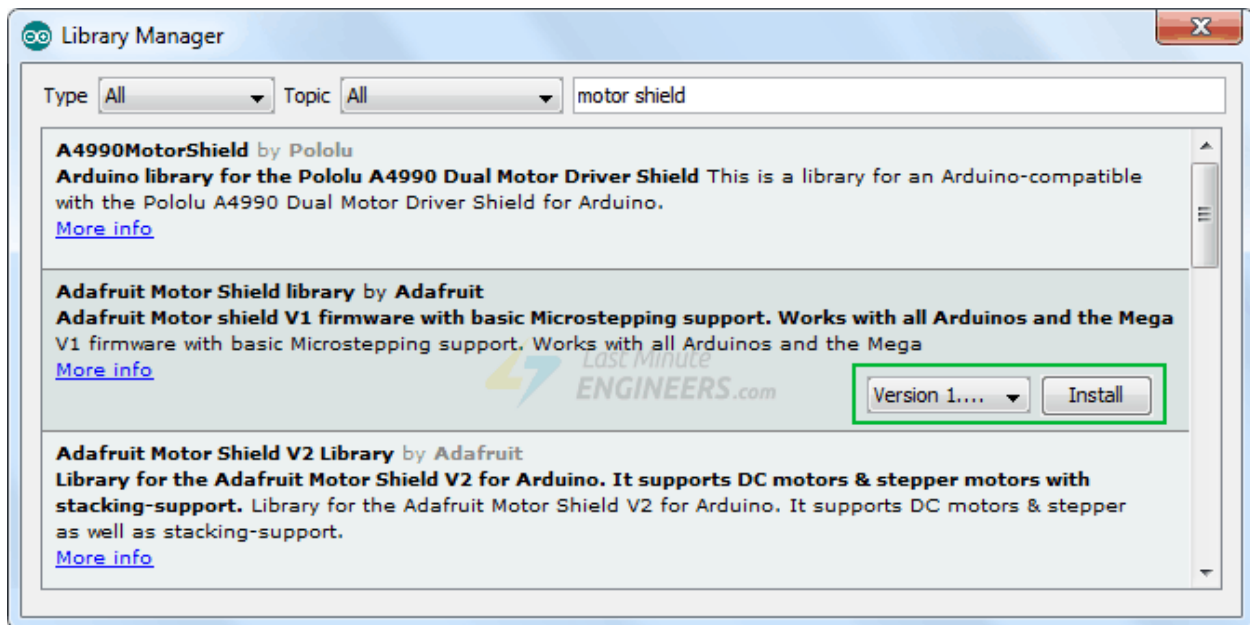
Installing AFMotor Library

In order to communicate with the shield, we need to install AFMotor.h library so that we can issue simple commands to control DC, stepper & servo motors.

To install the library navigate to the Sketch > Include Library > Manage Libraries... Wait for Library Manager to download libraries index and update list of installed libraries.



Filter your search by typing 'motor shield'. There should be a couple entries. Look for Adafruit Motor Shield library(V1 Firmware) by Adafruit. Click on that entry, and then select Install.



Control DC, Stepper & Servo with L293D

Motor Driver Shield & Arduino

If you are planning on assembling your new robot, you will eventually want to learn about controlling variety of motors like DC motors, Stepper motors & servos. One of the easiest and inexpensive way to do that is to interface L293D Motor Driver Shield with Arduino. It's a full-featured motor shield – perfect for many robot and CNC projects.

It can drive :

- 4 bi-directional DC motors with 8-bit speed selection(0-255)
- 2 stepper motors (unipolar or bipolar) with single coil, double coil, interleaved or micro-stepping.
- 2 servo motors

L293D Motor Driver & 74HC595 Shift Register



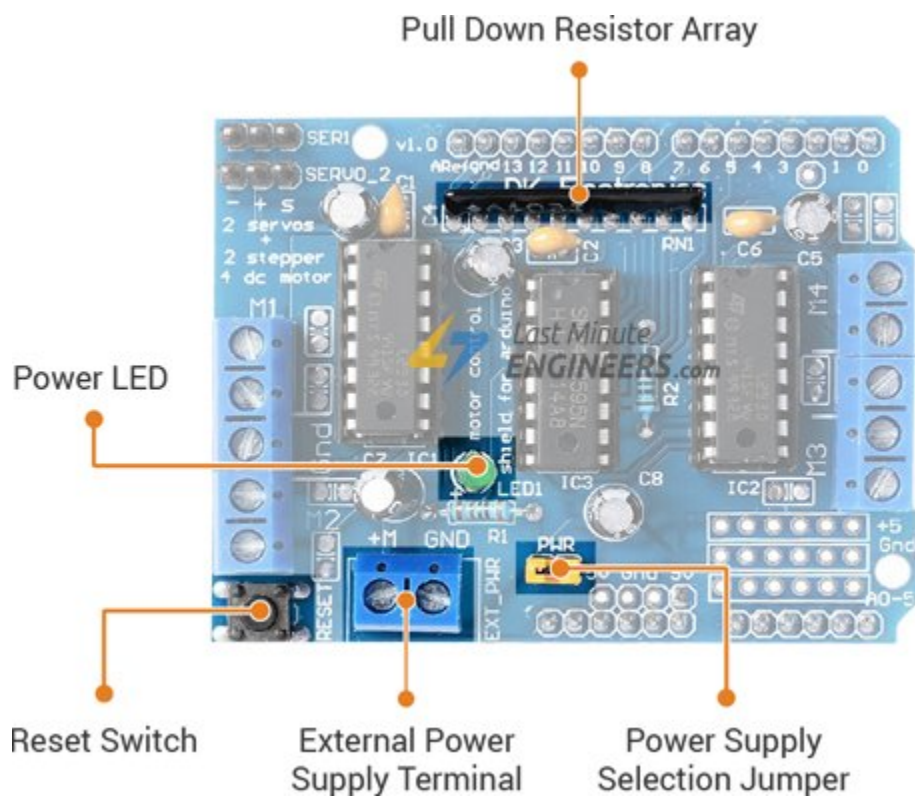
The L293D is a dual-channel H-Bridge motor driver capable of driving a pair of DC motors or single stepper motor.

As the shield comes with two L293D motor driver chipsets, that means it can individually drive up to four DC motors making it ideal for building four-wheel robot platforms.

The shield offers total 4 H-Bridges and each H-bridge can deliver up to 0.6A to the motor.

The shield also comes with a 74HC595 shift register that extends 4 digital pins of the Arduino to the 8 direction control pins of two L293D chips.

Power Supply



There exists three scenarios when it comes to supplying power for the motors through shield.

- Single DC power supply for both Arduino and motors: If you would like to have a single DC power supply for both Arduino and motors, simply plug it into the DC jack on the Arduino or the 2-pin EXT_PWR block on the shield. Place the power jumper on the motor shield. You can employ this method only when motor supply voltage is less than 12V.
- (Recommended) Arduino powered through USB and motors through a DC power supply: If you would like to have the Arduino powered off of USB and the motors powered off of a DC power supply, plug in the USB cable. Then connect the motor supply to the EXT_PWR block on the shield. Do not place the jumper on the shield.
- Two separate DC power supplies for the Arduino and motors: If you would like to have 2 separate DC power supplies for the Arduino and motors. Plug in the supply for the Arduino into the DC jack, and connect the motor supply to the EXT_PWR block. Make sure the jumper is removed from the motor shield.
- DO NOT supply power to the EXT_PWR input when jumper is in place. It may damage the motor shield and also your Arduino!

As a bonus, the shield offers below features:

- The shield comes with a pulldown resistor array to keep motors switched off during power-up.
- The on-board LED indicates the motor power supply is Okay. If it is not lit, the motors will not run.
- The RESET is nothing but Arduino's reset button. It just brought up top for convenience.

Output Terminals



The output channels of both the L293D chips are broken out to the edge of the shield with two 5-pin screw terminals viz. M1, M2, M3 & M4. You can connect four DC motors having voltages between 4.5 to 25V to these terminals. Each channel on the module can deliver up to 600mA to the DC motor. However, the amount of current supplied to the motor depends on system's power supply.

You can also connect two stepper motors to output terminals. One stepper motor to motor port M1-M2 and other to M3-M4. The GND terminal is also provided if you happen to have a unipolar stepper motor. You can connect the center taps of both stepper motors to this terminal.

The shield brings out the 16bit PWM output lines to two 3-pin headers to which you can connect two servo motors.

Unused pins on L293D shield

Digital pins #2, #13 and analog pins A0-A5 are not used by the shield.

Analog pins are broken out in the bottom right corner where pin 2 has a small breakout. If you want to use these pins, you can connect some headers to it.

Driving DC Motors with L293D Shield

Now that we know everything about the shield, we can begin hooking it up to our Arduino!

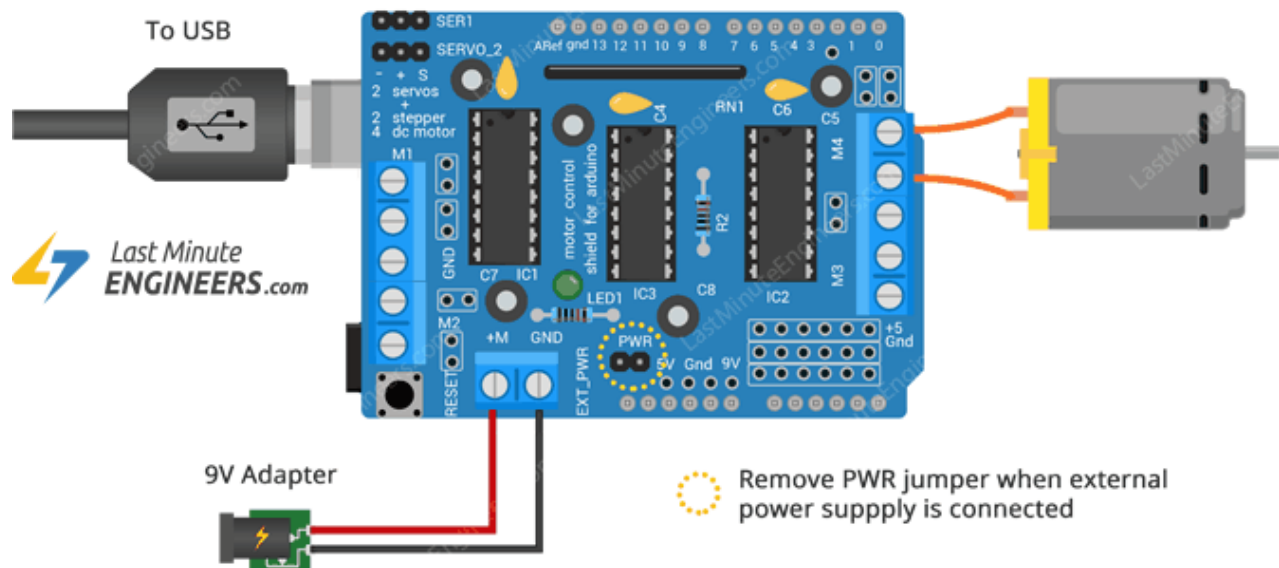
Start by plugging the shield on the top of the Arduino.

Next, connect power supply to the motors. Although you can connect DC motors having voltages between 4.5 to 25V to the shield, in our experiment we are using DC Motors that are rated for 9V. So, we will connect external 9V power supply to the EXT_PWR terminal.

Now, connect the motor to either M1, M2, M3 or M4 motor terminals. In our experiment we are connecting it to M4.

The following sketch will give you complete understanding on how to control speed and spinning direction of a DC motor with L293D motor driver shield and can serve as the basis for more practical experiments and projects.

Wiring-DC-Motor-to-L293D-Motor-Shield-Arduino



For Single DC Motor

```
#include <AFMotor.h>
```

```
AF_DCMotor motor(4);
```

```
void setup()
```

```
{  
    //Set initial speed of the motor & stop  
    motor.setSpeed(200);  
    motor.run(RELEASE);  
}
```

```
void loop()
```

```
{  
    uint8_t i;  
  
    // Turn on motor  
    motor.run(FORWARD);  
  
    // Accelerate from zero to maximum speed  
    for (i=0; i<255; i++)  
    {  
        motor.setSpeed(i);  
        delay(10);  
    }  
  
    // Decelerate from maximum speed to zero  
    for (i=255; i!=0; i--)
```

```
{  
    motor.setSpeed(i);  
    delay(10);  
}  
  
// Now change motor direction  
motor.run(BACKWARD);  
  
// Accelerate from zero to maximum speed  
for (i=0; i<255; i++)  
{  
    motor.setSpeed(i);  
    delay(10);  
}  
  
// Decelerate from maximum speed to zero  
for (i=255; i!=0; i--)  
{  
    motor.setSpeed(i);  
    delay(10);  
}  
  
// Now turn off motor  
motor.run(RELEASE);  
delay(1000);  
}
```

FOR TWO DC MOTORS

```
#include <AFMotor.h>
```

```
AF_DCMotor motor1(4);
```

```
AF_DCMotor motor2(3);
```

```
void setup()
```

```
{
```

```
  //Set initial speed of the motor & stop
```

```
  motor1.setSpeed(200);
```

```
  motor1.run(RELEASE);
```

```
  motor2.setSpeed(200);
```

```
  motor2.run(RELEASE);
```

```
}
```

```
void loop()
```

```
{
```

```
  uint8_t i;
```

```
  // Turn on motor
```

```
  motor1.run(FORWARD);
```

```
  motor2.run(FORWARD);
```

```
  // Accelerate from zero to maximum speed
```

```
  for (i=0; i<255; i++)
```

```
  {
```

```
    motor1.setSpeed(i);
```

```
    delay(10);
    motor2.setSpeed(i);
    delay(10);
}

// Decelerate from maximum speed to zero
for (i=255; i!=0; i--)
{
    motor1.setSpeed(i);
    delay(10);
    motor2.setSpeed(i);
    delay(10);
}

// Now change motor direction
motor1.run(BACKWARD);
motor2.run(BACKWARD);

// Accelerate from zero to maximum speed
for (i=0; i<255; i++)
{
    motor1.setSpeed(i);
    delay(10);
    motor2.setSpeed(i);
    delay(10);
}

// Decelerate from maximum speed to zero
for (i=255; i!=0; i--)
```

```
{  
    motor1.setSpeed(i);  
    delay(10);  
    motor2.setSpeed(i);  
    delay(10);  
}
```

```
// Now turn off motor  
motor1.run(RELEASE);  
delay(1000);  
motor2.run(RELEASE);  
delay(1000);  
}
```